

Reese Ludwick

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Summary

Applied AI engineer who ships LLM and agent applications to production end-to-end — RAG pipelines, multi-agent LangGraph systems, and custom MCP tool servers deployed for real clients. Strengths in applied AI system design, prompt engineering, and model/output evaluation, with hands-on delivery across FastAPI + Python backends, PostgreSQL/Redis, and async task pipelines. Focused on turning state-of-the-art LLM tooling into dependable, scalable software that works in front of real users.

Experience

Software Associate — Cognizant (*Remote, Dec 2025 – Present*) - Build GenAI and agentic AI systems with LangChain, LangGraph, and Python across R&D and client deliverables. - Design and ship RAG pipelines and MCP server integrations for production workflows. - Scope, prototype, and deliver AI data-analysis tools alongside cross-functional teams.

Freelance AI Engineer & Full-Stack Developer — WorldCovers / APMC (*Remote, May 2026 – Present*) - Paid contract engineer on production web properties and the WorldCovers platform. - Ship features and automation against a 14-table PostgreSQL schema using Django REST + React/TypeScript. - AI-augmented engineering (Claude Code) under compressed timelines, with direct stakeholder coordination.

GenAI Developer Contractor — Cognizant (*Remote, Jun – Dec 2025*) - Built GenAI POCs with LangChain, LangGraph, and Python. - Ran prompt engineering and model evaluation to optimize LLM outputs across business use cases. - Built custom FastMCP servers and external MCP integrations with AWS for dynamic agent tool access.

IT Student Helper — James Rumsey Technical Institute (*2019 – 2021*) - Resolved 300+ support tickets; led a Wi-Fi overhaul cutting latency 25%; completed 100+ AV/network setups.

Projects

CreatorClip / autoclip.studio — *Python, FastAPI, Celery, pgvector, WhisperX, Voyage AI, Anthropic API, ffmpeg, YouTube OAuth* Live SaaS (Stripe billing), shipped end-to-end. Async Celery workers run WhisperX transcription, Voyage + pgvector embeddings for semantic retrieval, and ffmpeg rendering. Custom creator-“DNA” module tailors output; YouTube OAuth for publishing. Production LLM/ML orchestration at scale; ~79% test coverage.

CFO Agent (*in active development*) — *Python, FastAPI, LangGraph, PostgreSQL, Redis, Anthropic API* FastAPI + LangGraph agent with PostgreSQL persistence and Redis caching over imported financial data (CSV/OFX import + encrypted vault). Hybrid context architecture; multi-tenant-ready design. Applied AI reasoning over financial data.

Agentic Software Development Team — *LangChain, LangGraph, Amazon Bedrock (Cognizant R&D)* Multi-agent LangGraph system (codegen / testing / security / supervision) automating the build lifecycle on Bedrock and OSS ML backends.

Customer Call Data Analysis & Insights Engine — *Python, NLP, Sentence Transformers (Cognizant R&D)* NLP clustering of call data into semantic issue groups; agentic AI for call consolidation and executive reporting.

Technical Skills

Languages: Python, SQL **AI / ML:** LangChain, LangGraph, RAG, MCP Servers, Amazon Bedrock, Sentence Transformers, TensorFlow, pgvector, Anthropic API, Voyage AI, WhisperX, prompt engineering, model/output evaluation **Backend / Infra:** FastAPI, Django REST, PostgreSQL, Redis, Celery, Docker **Frontend:** React / TypeScript **Data:** Pandas, NumPy, Jupyter **Tooling / Cloud:** Linux, Ubuntu, Git, GitHub, VS Code, Claude Code, Azure Fundamentals

Education

B.S. Computer Science — Shepherd University (*Aug 2021 – May 2025*) — Major GPA 3.43

Certifications

TensorFlow Developer Specialization (DeepLearning.ai) · Linux + Git (Boot.dev) · AZ-900 · GitHub Copilot · Generative AI Context Engineering (Cognizant)